

ISYE 6644 - SIMULATION: PANDEMIC FLU SPREAD

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1. Abstract

In this problem, a classroom of 51 kids were modeled to see how a disease would spread with the presence of one contagious kid who attended school for all of the 3 days he was sick. A Monte Carlo method was used in Python to simulate the experiment by running it 1000 times. The spread was observed under two conditions, one where all of the kids were susceptible and the other where each kid had a 60% chance of being immunized with the vaccines working perfectly. For the case without vaccination, the disease spread to around 18 kids running for around 20 days. But for the immunized scenario, the disease could spread to only about 2 kids and was quickly contained within 2-3 days. These findings indicate the importance of vaccines in containing contagious disease even in high contact situations.

2. Background

2.1 Literature Review

According to the paper by Kermack and McKendrick (1927), the problem being modeled is considered to be S-I-R. This states that for such diseases, individuals go through 3 different stages, initially they are considered to be susceptible (S) to catching the disease, once infected they go into the infectious (I) stage and finally moving to the recovered (R) stage. The paper also states that for such diseases, the spread depends more on the number of susceptible cases around rather than the number of infectious cases. This is evident in the research as well - with no immunization, the disease spreads to 18 individuals but when immunization is introduced, only 2 kids could be infected as the number of susceptible cases goes down.

Furthermore, the paper by Cauchemez et al. (2011) suggests that disease spread is 5 times more likely inside a classroom setting compared to other places. Their experiment in a classroom environment found disease to spread by 0.3-0.9 every week. This supports the finding of this report that disease reached its highest growth rate around day 20 indicating that it increased gradually rather than spiking instantly.

2.2 Description Of The Problem

For this project, Python was used to simulate the spread of an infectious disease in a classroom of 51 elementary kids. On the first day, a single student - Tommy (the first infectious case) interacted freely with the rest of the 50 kids in his classroom and exposed them to catching the flu.

Once infected, a kid was infectious for 3 days when he/she could spread the disease. But their infectious period only started from the next day leading to the total number of contagious cases on any given day being a sum of the newly infected cases in the last 3 days. Tommy also followed the same rule and attended school on all of the 3 days he was infected.

It was assumed that all kids and days were independently and identically distributed so each kid followed a Bernoulli trial with the chance of getting infected (p) being 0.01. Hence the total number of infections on each day followed a binomial distribution, $X \sim \text{Bin}(n, p)$ with n being the current number of healthy children - 50.

The simulation exercise aimed to study the spread of disease under two scenarios, given the scenario of high contact but low probability (only 0.01) of transmission.:

1. Conditions where none of the kids were immunized
2. Scenario with each kid having a 60% chance of being immunized, with the immunization being 100% effective

2.3 Methodology

The simulation was carried out in python based on the S-I-R logic suggested by Kermack and McKendrick (1927). Each of the kids were assigned a state among -1 to 4, with -1 being immunized (part E), 0 being healthy, 1-3: representing the day of infection, and lastly 4 representing students who already recovered. A Monte Carlo experiment was conducted 1000 times to carry out discrete stochastic simulation, with each day representing a state change and count of new diseases.

For each susceptible child, the probability to not get infected was calculated with $P(\text{not_getting_infected}) = 0.99^x$ where x was the number of infectious kids in the classroom. If this probability was less than a random generated number (between 0-1), the kid was considered to be infected and his status was changed from 0 to 1 to indicate his first day of infection on the next day.

The same logic was followed for part E (the immunized scenario), but in this case, for each susceptible child, a random generated number between 0 to 1 was compared against 0.60 to randomly assign an immunization shield to them.

2.4 Real World Application

Such experiments can help to significantly reduce disease spread especially in places like classrooms which act as a super spreader of diseases. Experimenting with different contagious windows and vaccination thresholds, simulations can be done to identify the ideal time frame for infectious kids to stay home to prevent the spread.

3. Main Findings

3.1 Model Verification

To ensure that the model was working properly, the model result from day 2 was compared to manual calculation to check if they were agreeing with each other. A Binomial distribution, $X \sim (n, p)$ with $n = 50$ and $p = 0.01$ suggests $50 * 0.01 = 0.5$ new infection on Day 2 on average. This result matches the output from our histogram which also indicates 1.5 infections by Day 2.

3.2 Solution

- (a) The simulation started with a total of 51 kids with only 1 kid - Tommy being infectious. He came into contact with the rest of the 50 kids for all of the 3 days he remained contagious. He had a 0.01 probability of infecting each kid so each individual interaction was a Bernoulli trial. But since he along with any other kid he infected had the same probability of infecting a kid, the combination of all days they interacted was treated as a Binomial distribution, $X \sim (50, 0.01)$.
- (b) Since the spread of disease followed a Binomial distribution, the number of kids expected to be infected by Tommy (the single infectious kid on day 1) is $50 * 0.01 = 0.5$ kids.
- (c) Following the answer from part B, the expected number of sick kids by day 2 is $0.5 + 1$ (Tommy), so 1.5 kids.
- (d) Scenario where kids were not immunized:

Figure 1.1 depicts the total expected number of kids sick by each day for the scenario where none of the kids were immunized. It clearly shows that it justifies the math that 1.5 kids are infected by day 2. Moreover, the flu peaks around day 20 and infects a total of 18 kids by then before starting to plateau.

In figure 1.2, the histogram for the duration of the epidemic shows a bimodal distribution where for around 350 simulations out of 1000, the infection started and stopped with Tommy being the only infected kid. This is because although he came into contact with 50 of his classmates, the probability of infecting a susceptible kid was only 0.01. For the rest of the simulations, when Tommy did manage to infect others, it can be observed that the infection peaks around Day 20 and eventually slows down towards 0 by day 40.

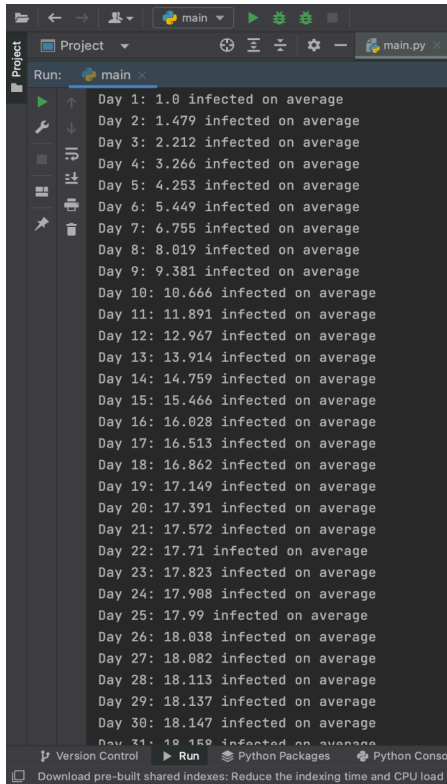


Fig 1.1 (left): Table representing the average cumulative infections on each day - no vaccination

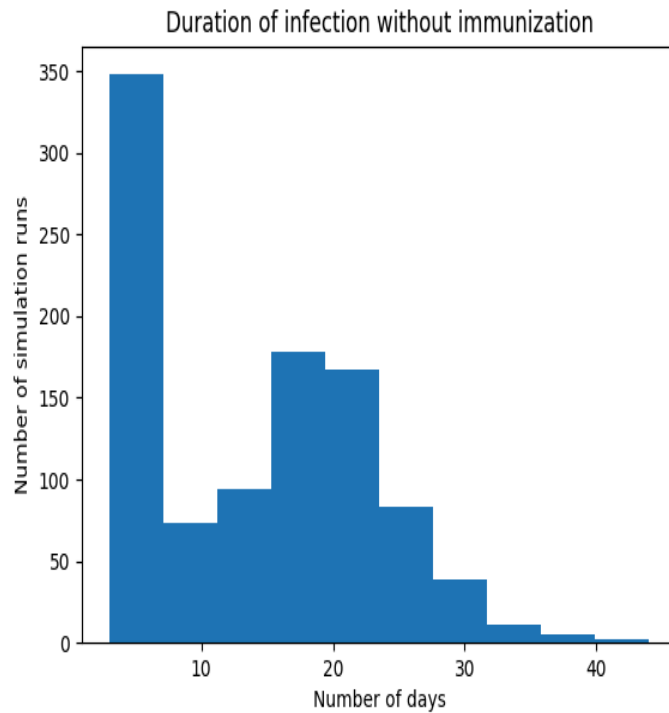


Fig1.2 (right): Histogram showing the duration of disease spread - no vaccination

(e) Scenario where each kid had a 60% chance of being immunized:

In figure 2.1, only around 2 kids were infected and the infection quickly plateaued by day 7. Figure 2.2 depicts that for about 600 simulations, the disease could not spread past Tommy. Even when it did manage to spread, it was quickly contained within 2 days. According to the data presented by the histogram, the curve is declining rapidly and shows that the disease was fully contained within 16 days.

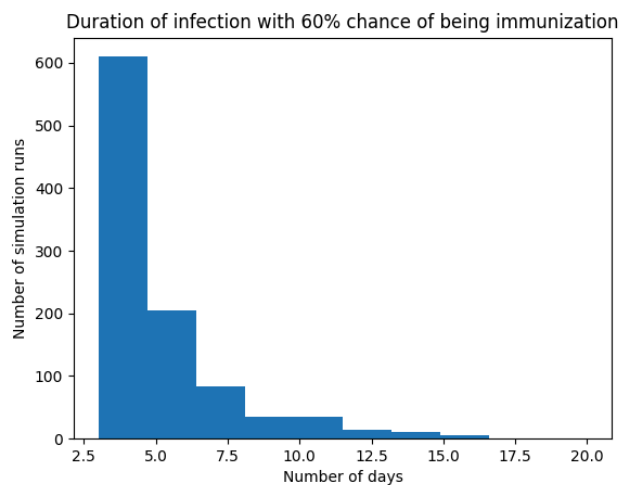
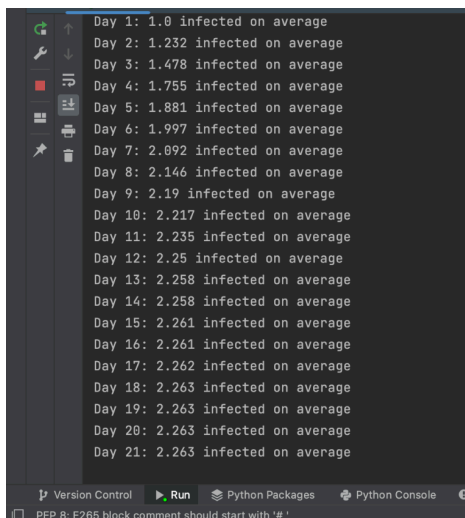


Fig 2.1 (left): Table representing the average cumulative infections on each day - with vaccination

Fig 2.2 (right): Histogram showing the duration of disease spread - with vaccination

4. Conclusions

Even though the probability of the disease spread was quite low at a probability of 0.01, it still managed to infect about 18 kids on average for the baseline scenario with no vaccinations indicating that even with low probabilities, a high contact environment can lead to substantial disease spread. In contrast, introductions to vaccines can help to break the transmission chain by introducing herd immunity where even if a group of the population wasn't vaccinated, the ones that were vaccinated, helped to break the spread of the disease.

In the future, a deeper analysis can be conducted by varying the recovery period from the current assumption of 3 days. Another area that can be introduced to make it more realistic could be removing the assumption that the vaccines work perfectly.

5. References & Citations

- [1] Kermack, W. O., & McKendrick, A. G. (1927). *A Contribution to the Mathematical Theory of Epidemics*. Proceedings of the Royal Society of London. Series A, 115(772), 700-721.
<https://jxshix.people.wm.edu/2009-harbin-course/classic/Kermack-McKendrick-1927-I.pdf>
- [2] Cauchemez, S., Bhattarai, A., Marchbanks, T. L., Fagan, R. P., Ostroff, S., Ferguson, N. M., ... & Finelli, L. (2011). *Role of social networks in shaping disease transmission during a community outbreak of 2009 H1N1 pandemic influenza*. Proceedings of the National Academy of Sciences, 108(7), 2825-2830. <https://doi.org/10.1073/pnas.1008895108>
- [3] Python Software Foundation. Python Language Reference, version 3.x. Available at <http://www.python.org>.
- [4] Gemini was used to help refine the disease state transition logic in Python while troubleshooting the calculation of cumulative average across the 1000 Monte Carlo experiments